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Lab01-01.exe

Submission name: Lab01-01.dll

Size: 160KiB

Type: pdf executable

Mime: application/x-dosexec

SHA256: f50e42c8dfaab649bde0398867e930b86c2a599e8db83b8260393082268f2dba

Submitted At: 2017-07-05 22:39:27 (UTC)

Last Anti-Virus Scan: 2026-02-24 14:56:19 (UTC)

Last Sandbox Report: 2024-05-30 02:10:31 (UTC)

malicious

Threat Score: 100/100

AV Detection: 77%

Labeled As: Win/malicious_confidence_100%

#tag

#nothing

#instascore

#upatre

#backdoor

#crypt

#downloader

#injector

View all tags (11)

X Post

Link

E-Mail

0

Community Score

0

Anti-Virus Results

Updated 3 months ago - Click to Refresh

CrowdStrike Falcon

Static Analysis and ML

Malicious (100%)

X No Additional Data

MetaDefender

Multi Scan Analysis

Malicious (14/26)

More Details

HYBRID ANALYSIS

Sandbox

Quick Scans

File Collections

Resources

Request Info

IP, Domain

Analysis Overview

Request Report Deletion

Show Sample Content

Submission name: Lab01-01.exe

Size: 16KiB

Type: pdf executable

Mime: application/x-dosexec

SHA256: 58898bd42c5bd3bf9b1389f0eee5b39cd59180e8370eb9ea838a0b327bd6fe47

Submitted At: 2017-07-05 22:14:28 (UTC)

Last Anti-Virus Scan: 2026-05-12 14:07:28 (UTC)

Last Sandbox Report: 2023-11-08 23:02:32 (UTC)

malicious

Threat Score: 100/100

AV Detection: 83%

Labeled As: Ulise.Generic

#mihi

#learning

#tag

#Lab01-01

#lab

#malwareanalysis

#1

X Post

Link

E-Mail

0

Community Score

0

Anti-Virus Results

Updated 19 days ago - Click to Refresh

CrowdStrike Falcon

Static Analysis and ML

Malicious (100%)

X No Additional Data

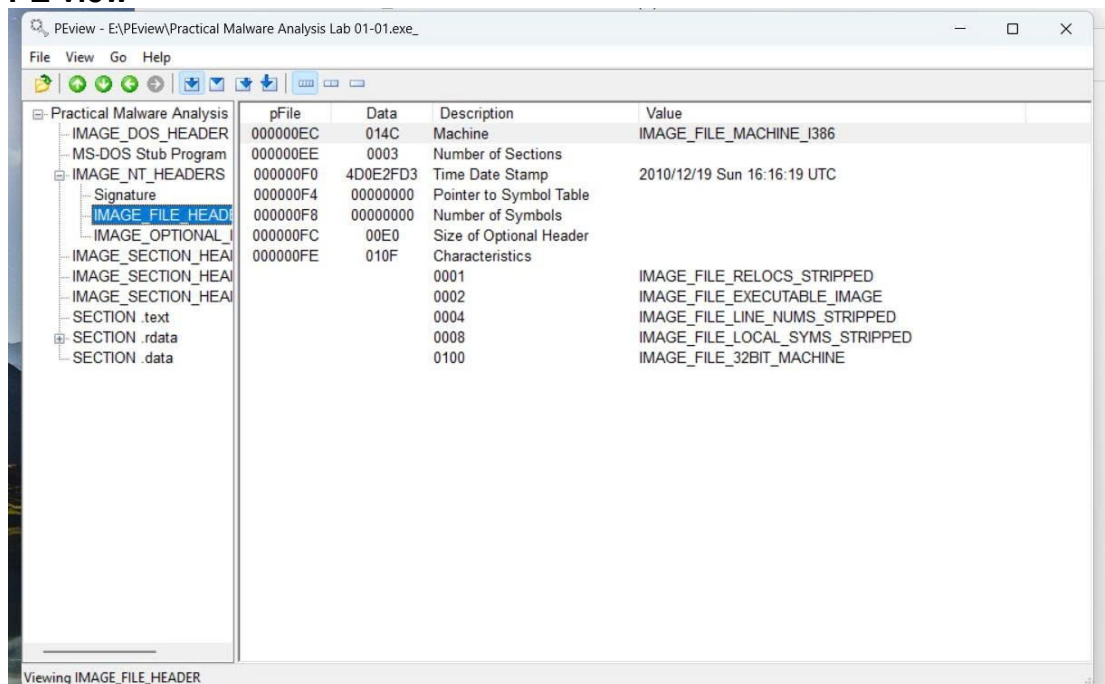
MetaDefender

Multi Scan Analysis

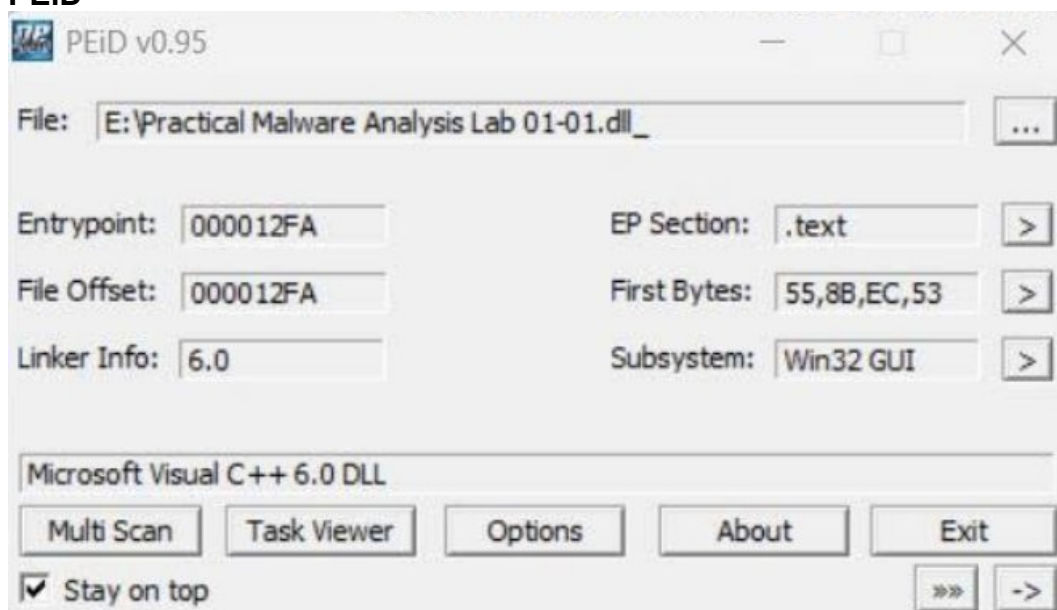
Malicious (17/26)

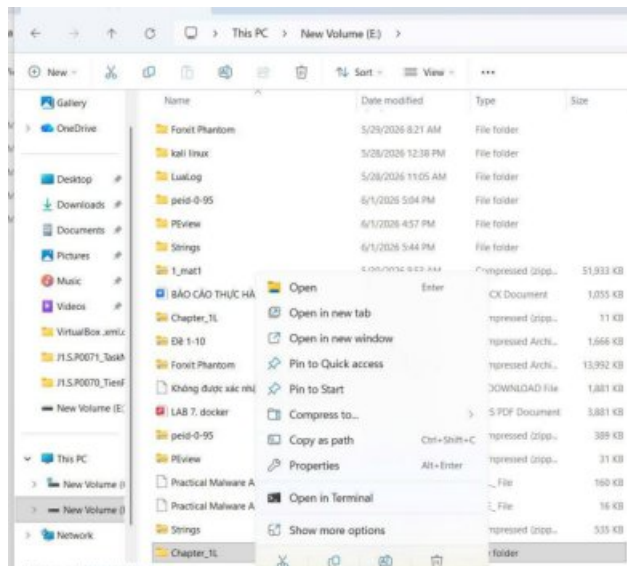
More Details

PE view



PEiD





strings Lab01-01.exe > str1exe.txt
notepad str1exe.txt

```
E:\Chapter_1L\Chapter_1L>strings Lab01-01.exe > strlex.txt
```

```
Strings v2.54 - Search for ANSI and Unicode strings in binary images.  
Copyright (C) 1999-2021 Mark Russinovich  
Sysinternals - www.sysinternals.com
```



Lab01-01.dll

```
NWVS
u7WPS
u&wVS
WVS
_ ^[]
%
closeHandle
Sleep
CreateProcessA
CreateMutexA
OpenMutexA
KERNEL32.dll
WS2_32.dll
strncmp
MSVCRT.dll
free
_initterm
malloc
_adjust_fdiv
exec
sleep
hello
127.26.152.13
SADFHUHF
/0I0[0h0p0
141G1[1l1
1Y2a2g2r2
3!3}3
```

Dependency Walker

Lab01-01.exe

LAB01-01.EXE

KERNEL32.DLL

API-MS-WIN-CORE-RTLSUPPORT-L1-1-0.DLL

API-MS-WIN-CORE-RTLSUPPORT-L1-2-0.DLL

NTDLL.DLL

KERNELBASE.DLL

API-MS-WIN-CORE-PROCESSTHREADS-L1-1-0.DLL

API-MS-WIN-CORE-PROCESSTHREADS-L1-1-1.DLL

API-MS-WIN-CORE-PROCESSTHREADS-L1-1-2.DLL

API-MS-WIN-CORE-PROCESSTHREADS-L1-1-3.DLL

API-MS-WIN-CORE-REGISTRY-L1-1-0.DLL

API-MS-WIN-CORE-HEAP-L1-1-0.DLL

API-MS-WIN-CORE-HEAP-L1-1-1.DLL

PI	Ordinal ^	Hint	Function	Entry Point
	N/A	148 (0x0094)	FindFirstFileA	Not Bound
	N/A	157 (0x009D)	FindNextFileA	Not Bound
	N/A	437 (0x1B5)	IsBadReadPtr	Not Bound
	N/A	470 (0x1D6)	MapViewOfFile	Not Bound
	N/A	688 (0x2B0)	UnmapViewOfFile	Not Bound

E	Ordinal ^	Hint	Function	Entry Point
	1 (0x0001)	74 (0x004A)	BaseThreadInitThunk	0x00015D30
	2 (0x0002)	927 (0x39F)	InterlockedPushListSList	NTDLL.RtlInterlockedPushListSList
	3 (0x0003)	1605 (0x645)	Wow64Transition	0x00082328
	4 (0x0004)	0 (0x0000)	AcquireSRWLockExclusive	NTDLL.RtlAcquireSRWLockExclusive
	5 (0x0005)	1 (0x0001)	AcquireSRWLockShared	NTDLL.RtlAcquireSRWLockShared

Module ^	File Time Stamp	Link Time Stamp	File Size	Attr.	Link Checksum	Real Checksum	CPU	Subsystem
ACLUUI.DLL	05/23/2026 10:39p	10/05/1984 7:00p	442,880	A	0x00078CAC	0x00078CAC	x86	GUI
ACTIVEDS.DLL	05/23/2026 10:39p	11/25/2102 11:33p	227,840	A	0x00042F6A	0x00042F6A	x86	Console
ADSLDPC.DLL	09/16/2025 2:41a	10/10/2032 5:25p	211,968	A	0x0003DC02	0x0003DC02	x86	Console
ADVAPI32.DLL	05/23/2026 10:39p	01/29/1986 5:49p	521,840	A	0x00080669	0x00080669	x86	Console
ADVPACK.DLL	05/23/2026 10:39p	02/01/2021 3:18a	124,928	A	0x000223E2	0x000223E2	x86	GUI

Dependency Walker - [Lab01-01]

File Edit View Options Profile Window Help

MSVCRT.DLL

API-MS-WIN-CORE-CONSOLE-L1-1-0.DLL

API-MS-WIN-CORE-CONSOLE-L1-2-0.DLL

API-MS-WIN-CORE-DATETIME-L1-1-0.DLL

API-MS-WIN-CORE-DEBUG-L1-1-0.DLL

API-MS-WIN-CORE-ERRORHANDLING-L1-1-0.DLL

API-MS-WIN-CORE-FIBERS-L1-1-0.DLL

API-MS-WIN-CORE-FILE-L1-1-0.DLL

API-MS-WIN-CORE-HANDLE-L1-1-0.DLL

PI	Ordinal ^	Hint	Function	Entry Point
	N/A	72 (0x0048)	_XcptFilter	Not Bound
	88 (0x0058)		_getmainargs	Not Bound
	N/A	100 (0x0064)	_p__initenv	Not Bound
	N/A	106 (0x006A)	_p__commode	Not Bound
	N/A	111 (0x006F)	_p__fmode	Not Bound

E	Ordinal ^	Hint	Function	Entry Point
	1 (0x0001)	1 (0x0001)	?_?non_rtti_object@@QAE@ABV0@@Z	0x000398
	2 (0x0002)	2 (0x0002)	?_?non_rtti_object@@QAE@PBD@Z	0x000398
	3 (0x0003)	3 (0x0003)	?_?bad_cast@@AAE@PBQBD@Z	0x000398
	4 (0x0004)	4 (0x0004)	?_?bad_cast@@QAE@ABQBD@Z	0x000398
	5 (0x0005)	5 (0x0005)	?_?bad_cast@@QAF@ARV0@@Z	0x000398

Module ^	File Time Stamp	Link Time Stamp	File Size	Attr.	Link Checksum	Real Checksum	CPU	Subsystem
IPPCOMMON.DLL	05/23/2026 10:39p	07/15/1984 10:31p	305,152	A	0x00056A1F	0x00056A1F	x86	Console
IRI.DLL	04/01/2024 2:22p	05/06/1974 10:48p	42,944	A	0x0001A36A	0x0001A36A	x86	Console
KERNEL32.DLL	05/23/2026 10:39p	04/08/2080 7:40p	683,336	A	0x000A7857	0x000A7857	x86	Console
KERNELBASE.DLL	05/23/2026 10:39p	11/17/2082 7:02p	2,920,960	A	0x002CFE2A	0x002CFE2A	x86	Console
KTMW32.DLL	04/01/2024 2:22p	04/01/1989 6:10p	21,504	A	0x0000B302	0x0000B302	x86	GUI

Lab01-01.dll

Dependency Walker - [Lab01-01.dll]

File Edit View Options Profile Window Help

NTDSAPI.DLL

FWPUCLNT.DLL

DHCPVC.DLL

DHCPVC6.DLL

EXT-MS-WIN-NETWORKING-WCMAPI

WS2_32.DLL

API-MS-WIN-CORE-COM-L1-1-0.DLL

EXT-MS-WIN-AUTHZ-CONTEXT-L1-1-0.DLL

API-MS-WIN-SECURITY-SDLL-L1-1-0.DLL

BCRYPTPRIMITIVES.DLL

WS2_32.DLL

MSVCRT.DLL

PI	Ordinal ^	Hint	Function	Entry Point
0x00000003	3 (0x0003)	N/A	N/A	Not Bound
0x00000004	4 (0x0004)	N/A	N/A	Not Bound
0x00000009	9 (0x0009)	N/A	N/A	Not Bound
0x00000011	11 (0x000B)	N/A	N/A	Not Bound
0x00000016	16 (0x0010)	N/A	N/A	Not Bound
0x00000019	19 (0x0013)	N/A	N/A	Not Bound

E	Ordinal ^	Hint	Function	Entry Point
0x00000001	1 (0x0001)	146 (0x0092)	accept	0x0003ADA0
0x00000002	2 (0x0002)	147 (0x0093)	bind	0x0000EE00
3 (0x0003)	148 (0x0094)	closesocket	0x00010200	
4 (0x0004)	149 (0x0095)	connect	0x00017830	
5 (0x0005)	156 (0x009C)	getpeername	0x00012FD0	
6 (0x0006)	161 (0x00A1)	getsockname	0x00011C50	

Module ^	File Time Stamp	Link Time Stamp	File Size	Attr.	Link Checksum	Real Checksum	CPU	Subsystem
DWMAPI.DLL	05/23/2026 10:39p	11/27/2008 11:27p	133,448	A	0x0002E58B	0x0002E58B	x86	GUI
DWRITE.DLL	05/23/2026 10:39p	07/31/2102 10:01p	2,129,920	A	0x00212BEA	0x00212BEA	x86	Console
DXGI.DLL	05/23/2026 10:39p	12/23/1981 7:14a	1,073,448	A	0x0011454A	0x0011454A	x86	Console
EAPPCFG.DLL	05/23/2026 10:39p	10/11/2092 11:54p	217,088	A	0x00043386	0x00043386	x86	Console
EAPPPRX.DLL	05/23/2026 10:39p	06/11/2082 11:41a	67,584	A	0x0001EA49	0x0001EA49	x86	Console

Error: At least one required implicit or forwarded dependency was not found.

Warning: At least one delay-load dependency module was not found.

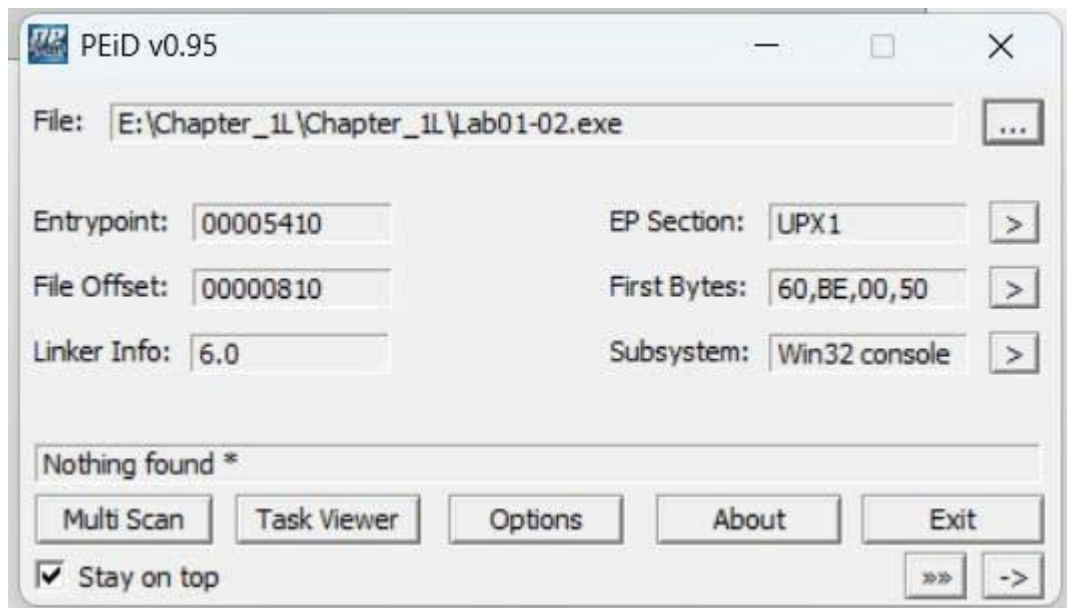
Warning: At least one module has an unresolved import due to a missing export function in a delay-load dependent module.

See Main report E1

Lab01-02.exe

Analysis Overview									
Request Report Deletion Show Sample Content Submission name: Lab01-02.exe Size: 3KiB Type: portable executable Mime: application/x-dosexec SHA256: c876a332d7d8da331cb8ee7ab7bf32752834d4b2b54eaa36267a2a48f6a6 Submitted At: 2017-07-05 22:28:42 (UTC) Last Anti-Virus Scan: 2025-10-25 10:51:51 (UTC) Last Sandbox Report: 2024-03-10 15:43:21 (UTC) malicious Threat Score: 100/100 AV Detection: 89% Labeled As: Ser/Us/Is/Generic #tag #Lab01-02 #adware #autorun #backdoor #crypt #downloader #exploit View all tags (15) Post Link Mail 0 Community Score 0 Analysis Overview Anti-Virus Scanner Results Falcon Sandbox Reports (10) Relations Incident Response Community (29) Back to top									
Anti-Virus Results									
Updated 7 months ago - Click to Refresh CrowdStrike Falcon Static Analysis and ML Malicious (100%) MetaDefender Multi Scan Analysis Malicious (21/27)									

PEiD



UPX -d -o Lab01-02-unpacked.exe Lab01-02.exe

```
E:\>cd Chapter_1L\Chapter_1L
E:\Chapter_1L\Chapter_1L>E:\upx-5.1.1-win64\upx-5.1.1-win64\upx.exe -d Lab01-02.exe -o Lab01-02_unpacked.exe
Ultimate Packer for eXecutables
Copyright (C) 1996 - 2026
UPX 5.1.1      Markus Oberhumer, Laszlo Molnar & John Reiser   Mar 5th 2026

  File size   Ratio   Format   Name
-----
  16384 <-   3072   18.75%   win32/pe   Lab01-02_unpacked.exe

Unpacked 1 file.
E:\Chapter_1L\Chapter_1L>
```

PEiD Lab01-02-unpacked.exe


```

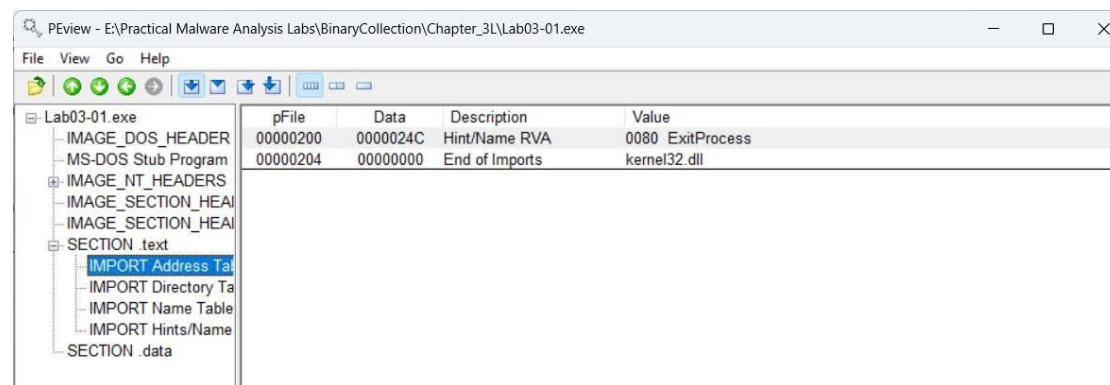
adjust_fdiv
p_commode
p_fmode
set_app_type
except_handler3
controlfp
InternetOpenUrlA
InternetOpenA

MalService
Malservice
HGL345
http://www.malwareanalysisbook.com Internet Explorer 8.0

```

1. Basic Dynamic Techniques

PEView Open Lab03-01.exe



Using Strings


```

QVlM
4~v
X:a
3sg
6I*h<8
^~m~m<|<|<|M
o/o/
00U
advapi32
ntdll
user32
Jbh
ww!
1+KY
x{w
#%li
}>*K
40j
QQVP
ucj
jjjjjj
advpack
hk7
~pj
<2f
StubPath
SOFTWARE\Classes\http\shell\open\commandv
Software\Microsoft\Active Setup\Installed Components\
test
www.practicalmalwareanalysis.com
admin
VideoDriver
WinVMX32-
vmx32to64.exe
SOFTWARE\Microsoft\Windows\CurrentVersion\Run

```

Run Process Explorer

Process Explorer - Sysinternals: www.sysinternals.com [DESKTOP-4M4AQ8K\Admin] (Administrator)

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	55,944 K	236		
Registry		9,392 K	64,904 K	280		
System Idle Process	90.89	60 K	8 K	0		
System	0.19	44 K	856 K	4		
Interrupts	0.57	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,252 K	1,688 K	824		
Memory Compression		40 K	0 K	3828		
csrss.exe		2,628 K	6,420 K	1116		
wininit.exe		1,920 K	9,672 K	1248		
services.exe		5,208 K	11,656 K	1324		
svchost.exe	< 0.01	11,340 K	40,600 K	1536	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		4,552 K	12,708 K	3376	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe	0.19	11,900 K	23,556 K	5640	WMI Provider Host	Microsoft Corporation
SearchHost.exe		45,596 K	126,988 K	5580		Microsoft Corporation
msedgewebview2.exe		43,232 K	124,816 K	7096	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,056 K	12,816 K	9420	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,604 K	42,140 K	10012	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		67,116 K	86,408 K	10036	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		10,368 K	21,196 K	10108	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		62,252 K	103,088 K	10372	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHost.exe		69,956 K	154,164 K	5620	Windows Start Experience H...	Microsoft Corporation
Widgets.exe		12,824 K	71,124 K	8228		Microsoft Corporation
msedgewebview2.exe		39,380 K	17,028 K	11452	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,012 K	13,472 K	1124	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		154,376 K	23,368 K	3392	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,184 K	8,620 K	3976	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		10,888 K	2,604 K	3756	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		115,316 K	4,912 K	11024	Microsoft Edge WebView2	Microsoft Corporation
RuntimeBroker.exe		6,868 K	43,944 K	8372	Runtime Broker	Microsoft Corporation
ShellExperienceHost.exe	Susp...	21,304 K	83,572 K	9120	Windows Shell Experience H...	Microsoft Corporation

CPU Usage: 8.78% Commit Charge: 35.30% Processes: 186 Physical Usage: 37.66%

Start Process Monitor

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKCU	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKCU	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKCU\Software\Classes\Local Settings\...	REPARSE	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKCU\Software\Classes\Local Settings\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKCU	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegQueryValue	HKCU\Software\Classes\Local Settings\...	BUFFER OVERFL...	Length: 12
9:23:17...	Explorer.EXE	7592	RegQueryValue	HKCU\Software\Classes\Local Settings\...	BUFFER OVERFL...	Length: 144
9:23:17...	Explorer.EXE	7592	RegQueryValue	HKCU\Software\Classes\Local Settings\...	SUCCESS	Type: REG_SZ, Le...
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\Software\Microsoft\Windows\Curr...	NAME NOT FOUND	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	KeySetInformation...
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegQueryValue	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegEnumKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Index: 0, Name: 130
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegQueryValue	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	
9:23:17...	Explorer.EXE	7592	RegQueryKey	HKLM	SUCCESS	Query: Handle Tag...
9:23:17...	Explorer.EXE	7592	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	Desired Access: R...
9:23:17...	Explorer.EXE	7592	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\Windows\...	SUCCESS	KeySetInformation...

Showing 384,049 of 687,321 events (55%) Backed by virtual memory

Excluding Harmless Processes

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time of Day	Process Name	PID	Operation	Path	Result	Detail
10:13:09 4601445 PM	lsass.exe	1352	QueryNameInfo...	C:\Users\Admin\AppData\Local\Temp\...	SUCCESS	Name: \Users\Adm...
10:13:09 4602168 PM	lsass.exe			Data\Local\Temp\...	SUCCESS	Name: \Users\Adm...
10:13:09 4609506 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4609740 PM	ctfmon.exe			soft\input\Locales\...	SUCCESS	Desired Access: R...
10:13:09 4609971 PM	ctfmon.exe			ficrosoft\input\Loca...	SUCCESS	Type: REG_DWO...
10:13:09 4610145 PM	ctfmon.exe			ficrosoft\input\Loca...	SUCCESS	
10:13:09 4611954 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Query: Handle Tag...
10:13:09 4612099 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Desired Access: R...
10:13:09 4612355 PM	ctfmon.exe			soft\input\Settings\...	NAME NOT FOUND	Length: 16
10:13:09 4612497 PM	ctfmon.exe			soft\input\Settings	SUCCESS	
10:13:09 4612616 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4612744 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Desired Access: R...
10:13:09 4612893 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Type: REG_DWO...
10:13:09 4613028 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	
10:13:09 4613132 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4613263 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	Desired Access: R...
10:13:09 4613409 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	Type: REG_BINAR...
10:13:09 4613539 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	
10:13:09 4613789 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4613932 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	Desired Access: W...
10:13:09 4614195 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	Type: REG_BINAR...
10:13:09 4614521 PM	ctfmon.exe			soft\input\TypingIn...	SUCCESS	
10:13:09 4614647 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4614763 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Desired Access: R...
10:13:09 4614887 PM	ctfmon.exe			soft\input\Settings\...	NAME NOT FOUND	Length: 16
10:13:09 4614998 PM	ctfmon.exe			soft\input\Settings	SUCCESS	
10:13:09 4615104 PM	ctfmon.exe				SUCCESS	Query: Handle Tag...
10:13:09 4615212 PM	ctfmon.exe			ficrosoft\input\Setti...	SUCCESS	Desired Access: R...
10:13:09 4615329 PM	ctfmon.exe				NAME NOT FOUND	Length: 16
10:13:09 4615451 PM	ctfmon.exe	11240	RegQueryValue	HKLM\SOFTWARE\Microsoft\input\Setti...	SUCCESS	
10:13:09 4615587 PM	ctfmon.exe	11240	RegCloseKey	HKLM\SOFTWARE\Microsoft\input\Setti...	SUCCESS	
10:13:09 4615687 PM	ctfmon.exe	11240	RegQueryKey	HKLM	SUCCESS	Query: Handle Tag...
10:13:09 4616859 PM	ctfmon.exe	11240	RegOpenKey	HKLM\Software\Microsoft\input\Locales\...	SUCCESS	Desired Access: R...

Properties... **Ctrl + P**

Stack... **Ctrl + K**

Toggle Bookmark **Ctrl + B**

Jump To... **Ctrl + J**

Count Occurrences...

Search Online...

Include 'lsass.exe'

Exclude 'lsass.exe'

Highlight 'lsass.exe'

Copy 'lsass.exe'

Edit Filter 'lsass.exe'

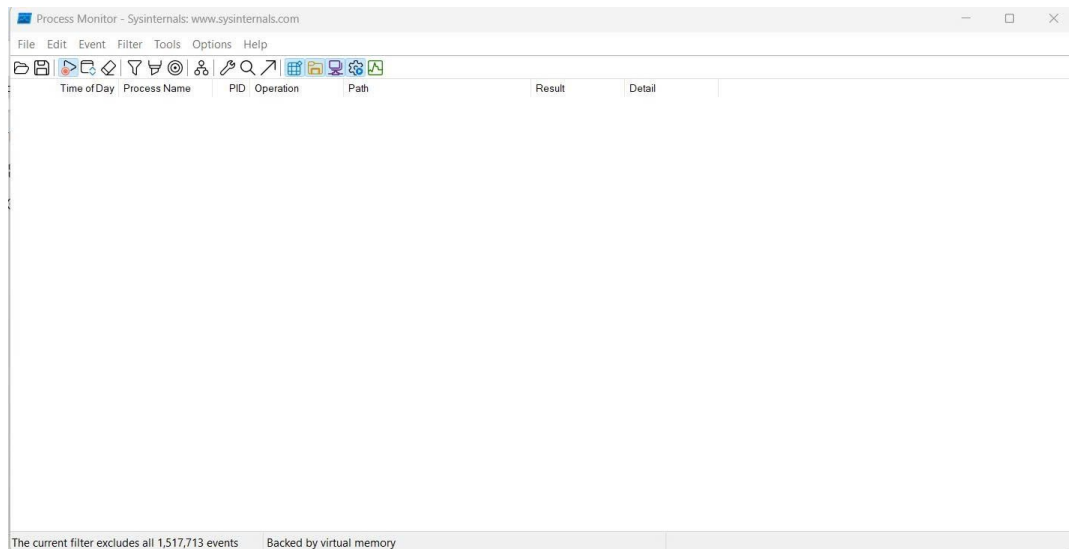
Exclude Events Before

Exclude Events After

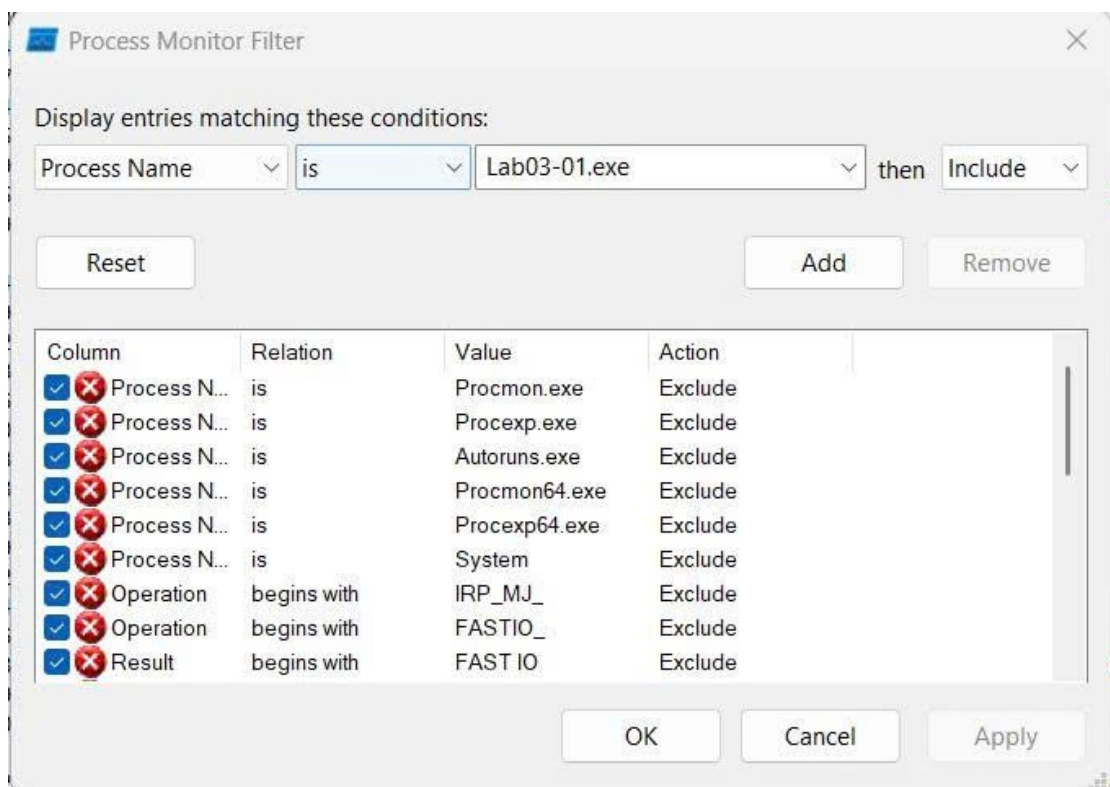
Include >

Exclude >

Highlight >



Viewing the Malicious Process's Events in Process Monitor



Viewing INetSim Logs


```
kali@kali: ~  
Session Actions Edit View Help  
* smtps_465_tcp - stopped (PID 5204)  
* http_80_tcp - stopped (PID 5201)  
* https_443_tcp - stopped (PID 5202)  
* dummy_1_udp - stopped (PID 5228)  
* dummy_1_tcp - stopped (PID 5227)  
* quotd_17_udp - stopped (PID 5224)  
* echo_7_udp - stopped (PID 5220)  
* ntp_123_udp - stopped (PID 5211)  
* daytime_13_tcp - stopped (PID 5217)  
* finger_79_tcp - stopped (PID 5212)  
* pop3s_995_tcp - stopped (PID 5206)  
* ftp_21_tcp - stopped (PID 5207)  
* syslog_514_udp - stopped (PID 5214)  
* http_80_tcp - stopped (PID 5201)  
* smtp_25_tcp - stopped (PID 5203)  
* time_37_tcp - stopped (PID 5215)  
* daytime_13_udp - stopped (PID 5218)  
* chargen_19_tcp - stopped (PID 5225)  
* quotd_17_tcp - stopped (PID 5223)  
* chargen_19_udp - stopped (PID 5226)  
* discard_9_udp - stopped (PID 5222)  
* tftp_69_udp - stopped (PID 5209)  
* irc_6667_tcp - stopped (PID 5210)  
Simulation stopped.  
== INetSim main process stopped (PID 5198) ==  
.  
--(kali@kali)-[~]
```

nano /var/log/inetsim/report/report.3384.txt

```
kali@kali: ~  
Session Actions Edit View Help  
GNU nano 8.7.1 /var/log/inetsim/report/report.12345.txt *  
2024-04-16 14:47:46 HTTP connection, method: GET, URL: http://kali.com/favico>  
2024-04-16 14:47:47 DNS connection, type: A, class: IN, requested name: dete>  
2024-04-16 14:47:47 DNS connection, type: A, class: IN, requested name: dete>  
2024-04-16 14:47:47 HTTP connection, method: GET, URL: http://detectportal.f>  
2024-04-16 14:47:50 DNS connection, type: A, class: IN, requested name: dete>  
2024-04-16 14:47:50 HTTP connection, method: GET, URL: http://detectportal.f>  
2024-04-16 14:47:53 DNS connection, type: A, class: IN, requested name: dete>  
2024-04-16 14:47:53 HTTP connection, method: GET, URL: http://detectportal.f>  
2024-04-16 14:50:36 DNS connection, type: A, class: IN, requested name: www.>  
2024-04-16 14:52:05 DNS connection, type: A, class: IN, requested name: time>  
2024-04-16 14:52:05 NTP connection, time received: 1713304159, time sent: 17>  
2024-04-16 14:52:07 DNS connection, type: PTR, class: IN, requested name: 13>  
2024-04-16 14:53:15 DNS connection, type: PTR, class: IN, requested name: 2.>  
2024-04-16 14:56:45 DNS connection, type: PTR, class: IN, requested name: 25>  
2024-04-16 14:57:40 DNS connection, type: ANY, class: IN, requested name: wp>  
2024-04-16 14:59:20 DNS connection, type: ANY, class: IN, requested name: wp>  
2024-04-16 15:01:20 DNS connection, type: ANY, class: IN, requested name: wp>  
2024-04-16 15:01:48 DNS connection, type: A, class: IN, requested name: www.>  
2024-04-16 15:05:42 DNS connection, type: PTR, class: IN, requested name: e.>  
2024-04-16 15:05:42 DNS connection, type: PTR, class: IN, requested name: 3.>  
2024-04-16 15:05:42 DNS connection, type: PTR, class: IN, requested name: 1.>  
2024-04-16 15:05:42 DNS connection, type: PTR, class: IN, requested name: 25>  
2024-04-16 15:05:42 Last simulated date in log file  
[ ^W = Ctrl+W M-W = Alt+W ]  
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
```

Bài tập Crackme 6

Em open file in x32dbg, to Strings References để xác định vị trí thông báo lỗi khi khởi động.

Em tìm thấy đoạn code push các tham số rồi gọi MessageBox.

Để loại bỏ nó, em sẽ thay đổi toàn bộ đoạn đó bằng lệnh nop.

004010A7	6A 40	push 40	
004010A9	68 61304000	push target.403061	403061: "[NAG] Please register this
004010AE	68 86304000	push target.403086	403086: "[BULLSHIT] Please register
004010B3	FF75 08	push dword ptr ss:[ebp+8]	
004010B6	FF75 08	call <JMP.&MessageBoxA>	

Em muốn khi nhấn Kiểm tra lại thì hiện thông báo "Cảm ơn bạn đã đăng ký phần mềm này".

Em vào String References để tìm câu đó.

Đồng thời em muốn hiểu tại sao nó lại hiện "Nag không bị xóa" thay vì "Clean crack!

Good Job!" ở dòng Trạng thái.

803D B0324000 02	cmp byte ptr ds:[4032B0],4	
74 1A	je target.401117	
803D B0324000 01	cmp byte ptr ds:[4032B0],1	
74 22	je target.401128	
68 21304000	push target.403021	403021: "Dirty crack! Nag r
6A 73	push 73	
FF75 08	push dword ptr ss:[ebp+8]	
E8 FB000000	call <JMP.&SetDlgItemTextA>	
EB 36	jmp target.40114D	
68 10304000	push target.403010	403010: "Nag not removed!"
6A 73	push 73	
FF75 08	push dword ptr ss:[ebp+8]	
E8 EA000000	call <JMP.&SetDlgItemTextA>	
EB 25	jmp target.40114D	
68 4A304000	push target.40304A	40304A: "clean crack! Good
6A 73	push 73	
FF75 08	push dword ptr ss:[ebp+8]	

Em xóa điều kiện nhảy đến "Nag not Remove" bằng cách đổi byte 74 1A thành nop.

Keep back line je to "Clean crack! Good Job!" chọn trường hợp bằng 1.

Như vậy khi bằng 2, chương trình không còn nhảy sai nữa mà chạy xuống đúng chỗ em mong muốn.

```

803D B0324000 02 cmp byte ptr ds:[4032B0],2
90 nop
90 nop
90 nop
90 nop
90 nop
90 nop
90 nop
90 nop
90 nop
90 nop
74 22 je target.401128
68 21304000 push target.403021
6A 73 push 73
FF75 08 push dword ptr ss:[ebp+8]
E8 FB000000 call <JMP.&SetDlgItemTextA>
EB 36 jmp target.40114D
68 10304000 push target.403010
6A 73 push 73
FF75 08 push dword ptr ss:[ebp+8]
E8 EA000000 call <JMP.&SetDlgItemTextA>
EB 25 jmp target.40114D
68 4A304000 push target.40304A
6A 73 push 73
FF75 08 push dword ptr ss:[ebp+8]

```

Khi nhảy đến phòng "Clean crack! Good Job!", em thấy nó tự động hiện ra "Thank you for posted this software".

Nguyên nhân là vì trong đoạn mã cũ có lệnh jmp nhảy thẳng qua cái đó.

EB 25	jmp target.40114D	
68 4A304000	push target.40304A	40304A:"Clean crack! Good
6A 73	push 73	
FF75 08	push dword ptr ss:[ebp+8]	
E8 D9000000	call <JMP.&SetDlgItemTextA>	
6A 40	push 40	
68 6E324000	push target.40326E	40326E:"Thank you!"
68 79324000	push target.403279	403279:"Thank you for regi
FF75 08	push dword ptr ss:[ebp+8]	
E8 B9000000	call <JMP.&MessageBoxA>	
EB 00	jmp target.40114D	

DONE!

